Heterogeneous Feature Collaboration Network for Salient Object Detection in Optical Remote Sensing Images

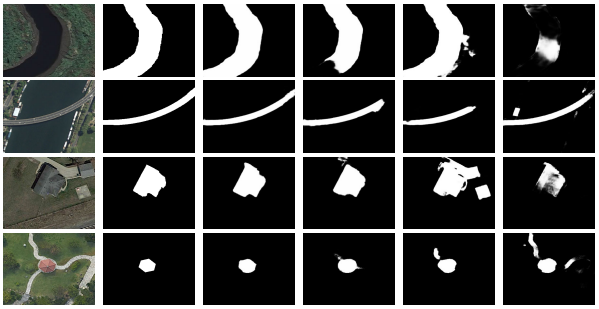
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***Abstract*— In recent years, the task of salient object detection in optical remote sensing images (RSI-SOD) has gained increasing interest. Despite some advancement in current methods, challenges such as the irregular topology of salient objects and cluttered backgrounds in optical RSI remain. To tackle these issues, we propose a novel heterogeneous feature collaboration network (HFCNet). Specifically, we design a new hybrid heterogeneous encoder that combines CNN and transformer to extract a set of heterogeneous features, famous in modeling local and global information, respectively. Subsequently, the adaptive global–local integration (AGLI) module is devised to integrate these complementary heterogeneous features through our feature alignment methods at global and local levels, so the global irregular topology structure and local details can be well-modeled. Furthermore, the proposed saliency-guided attention enhanced decoder (SGAED) leverages deep salient cues to guide the shallow decoders to pay more attention to important areas and suppress irrelevant areas, reducing the interference of cluttered backgrounds. Extensive experiments on three benchmark datasets have confirmed the significant superiority of our method compared with 18 state-of-the-art methods. All codes and results of our method are available at https://github.com/xumingzhu989/HFCNet-TGRS.**

***Index Terms*— Attention mechanism, feature alignment, heterogeneous feature fusion, optical remote sensing images, salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) is a critical task aimed at identifying the most visually appealing objects or regions within a given visual scene. In recent years, the rapid development of deep learning techniques has made SOD in natural scene images (NSI-SOD) [1], [2], [3], [4], [5] a great success. It has been successfully applied in various vision tasks, such as image/video segmentation [6], [7], [8], surface defect detection [9], [10], image quality assessment [11], [12], and visual tracking [13]. Meanwhile, SOD in optical remote sensing images (RSI-SOD) has received increasing attention. The goal of RSI-SOD is to detect attractive objects in optical RSI, such as aircraft, vehicles, ships, rivers, and islands. It plays an important role in environmental change monitoring, disaster detection, military reconnaissance, etc. Different from NSI captured by photographers with handheld cameras, optical RSI refers to color images captured by satellites or aerial sensors with wavelengths in the range of 400–760 nm, which results in a significant gap between NSI-SOD and RSI-SOD. Therefore, directly applying mature NSI-SOD solutions to RSI-SOD often results in unsatisfactory performance [as shown in Fig. 1(f)]. This article explores a specialized solution for RSI-SOD aimed at addressing the unique challenging issues in optical RSI.



(a) (b) (c) (d) (e) (f)

Fig. 1. Performance comparison between our method and the advanced RSI-SOD and NSI-SOD methods in classic challenging scenarios. (a) Optical RSI. (b) Ground truth. (c) Our method. (d) AESINet (RSI-SOD method). (e) MCCNet (RSI-SOD method). (f) MINet (NSI-SOD method).

Specifically, there are several challenges in processing optical RSI. First, salient objects in optical RSI typically exhibit irregular topology (as shown in the first and second rows of Fig. 1). For irregular topology, global semantic information helps in locating salient targets, while local detail information plays a key role in depicting the spatial details of these targets. Therefore, it is essential to thoroughly explore both global and local information to enhance the representation of irregular topology. The existing CNN-based models are capable of capturing local spatial details but are limited in learning global representations. In contrast, transformer-based models can easily establish long-range dependencies but lack

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the inductive bias inherent in CNN to model local context. Previous CNN-based methods [14], [15] use dilated convolutions or pooling operations to expand the receptive field. However, due to inherent translational invariance of CNN, they are unable to establish semantic relationships among distinct spatial positions. Notably, Wang et al. [16] propose a hybrid model combining CNN and transformer. It uses CNN blocks for effective local context modeling at shallow layers and deploys transformer blocks at deep layers for long-range context learning. However, relying solely on transformer blocks for feature extraction at deep layers may result in the loss of local detail information. To address this issue, we design a hybrid heterogeneous encoder that effectively combines the strengths of both CNN and transformer. This encoder uses CNN for local context modeling and incorporates transformer at deep layers to assist in establishing long-range semantic relationships among distinct spatial positions in the image. The two heterogeneous features extracted from the image by CNN and transformer exhibit rich complementarity, yet their significant differences pose a challenge for direct fusion. To effectively merge these heterogeneous features, we introduce the adaptive global–local integration (AGLI) module. The AGLI module progressively aligns the two heterogeneous features at both global semantic and local detail levels, effectively narrowing the semantic gap between them. Ultimately, these two heterogeneous features are integrated to obtain a more effective data representation that combines both global semantic information and local detail information.

Second, optical RSIs generally have cluttered backgrounds (as shown in the third and fourth rows of Fig. 1). Since optical RSIs usually contain various natural landforms and man-made structures, these images often display a cluttered background. In addition, the acquisition process of optical RSI is susceptible to factors such as lighting intensity and weather conditions. Consequently, salient objects in optical RSI may be accompanied by shadows, overexposure, and other disturbances, making it difficult to distinguish these objects from the background. Previous methods [17], [18] have used attention mechanisms to suppress background interference. However, in these methods, attention maps are usually generated independently for each feature layer, neglecting the interactions and complementarity between features at different layers. To tackle this issue, we propose a novel saliency-guided attention enhanced decoder (SGAED). Within this module, deep salient cues are transmitted to shallow decoders, guiding the generation of shallow attention maps in both the channel and spatial dimensions. This process effectively filters out redundant background interference and highlights salient objects in the shallow representations.

Our contributions are summarized as follows.

1) We propose a novel heterogeneous feature collaboration network (HFCNet) for RSI-SOD. The features generated from CNN and transformer can be integrated in our feature alignment module adaptively, and the deep salient cues are introduced to shallow decoders to exclude background interference progressively.
2) We propose a novel AGLI module to perceive the irregular topology of salient objects. It attempts to fuse complementary heterogeneous features by aligning global and local features, and then the global irregular topology structure and local details can be well-modeled, simultaneously.
3) We propose a novel SGAED to mitigate the effects of cluttered backgrounds. It propagates deep salient cues to shallow attention units, guiding the decoder to pay more attention to important areas and suppress irrelevant cluttered background regions.
4) We conduct comprehensive experiments on three challenging benchmark datasets. A series of ablation experiments have validated the effectiveness of different components in our model, and extensive experiments show the superiority of our model against 18 state-of-the-art methods.

II. Related Works

In this section, we will discuss the most related works to ours, including NSI-SOD and RSI-SOD.

A. Salient Object Detection for Natural Scene Images

In early research, NSI-SOD primarily relied on hand-crafted-based methods [19], [20]. These methods are based on features designed by experts and heuristic rules derived from experience. Due to the subjectivity of design and limited applicability, hand-crafted-based methods often failed to accurately identify salient objects in images. Thereafter, researchers started using traditional machine learning algorithms to design saliency models, which greatly improved the performance of these models. For example, Zhou et al. [21] constructed a two-layer sparse graph. Yuan et al. [2] proposed a regularized random walk ranking model. Peng et al. [22] designed a structured matrix factorization model with two structural regularizations. Xu et al. [23], [24] used graph clustering algorithms to discover the salient objects.

With the rapid development of deep learning technologies, many saliency models based on deep learning have been designed. To model global context, researchers have explored various methods, including attention mechanism [25], extended pooling operation [26], recurrent network [27], [28], graph neural network [29], [30], and dilated convolutions [5]. For example, Li et al. [25] embedded channel attention into the decoder to deliver cross-level complementary information. Liu et al. [26] used the global context information to guide feature decoding. Deng et al. [27] used a recurrent residual refinement network that can alternately integrate shallow and deep features. Xu et al. [29] introduced an innovative edge-gated graph convolution operation, leveraging feature similarity and spatial distance to accurately assess neighbor confidence and aggregate saliency information weighted by neighbor importance. Zhao et al. [5] designed a Fold-ASPP module to capture richer context information and localize salient objects of various sizes. To model local context, many methods have been adopted, including edge detection [3], [31] and multiscale feature fusion [4], [32], [33]. Specifically, Zhao et al. [3] proposed an EGNet to model complementary salient object information and salient edge information.

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Pang et al. [4] integrated features from adjacent levels and then used self-interaction modules to obtain multiscale features.

These NSI-SOD methods have achieved significant progress. However, the unique characteristics of optical RSI in terms of object types, object scales, and imaging direction make it challenging to directly apply NSI-SOD methods to RSI-SOD. Nevertheless, these NSI-SOD methods provide valuable inspiration for further exploration in RSI-SOD.

B. Salient Object Detection in Optical Remote Sensing Images

Despite the remarkable performance achieved by the NSI-SOD methods, research on RSI-SOD is relatively limited. In the early stages, many excellent researchers had proposed some traditional methods for RSI-SOD. For example, Zhang et al. [34] proposed a two-way complementary saliency analysis approach that combines vision-oriented saliency and knowledge-oriented saliency. Zhang et al. [35] designed a novel method based on dual-tree complex wavelet transform for RSI to generate texture feature pyramids. Liu et al. [36] used the CIE Lab space to build the Color Markov Chain and generate a bottom–up latent saliency map.

In recent years, deep learning techniques have also been applied to RSI-SOD model. Li et al. [37] constructed the first publicly available optical RSI dataset (i.e., ORSSD) and proposed a two-stream pyramid module and an encoder–decoder module with nested connections. Later, Zhang et al. [17] extended ORSSD and built a larger scale and challenging optical RSI dataset (i.e., EORSSD). They further proposed a dense attention fluid network to capture the global context. Li et al. [18] explored the complementarity of multiple content in features of optical RSI. Tu et al. [38] unveiled a more comprehensive RSI dataset (i.e., ORSI4199). They creatively devised a multiscale joint boundary and region model. Cong et al. [39] used a parallel multiscale attention mechanism to construct a relational inference model for RSI-SOD. Gao et al. [40] designed an adaptive spatial tokenization transformer encoder to extract global–local features. Zhou et al. [41] used edge information to generate target location cues at each decoding step. Zeng et al. [14] proposed a difference perception mechanism to adaptively enhance edge information of salient targets.

Despite the promising progress achieved by the above methods, challenges faced by optical RSI are still not fully resolved, such as irregular topology structure of salient objects and interference from complex backgrounds. To address these issues, we propose a novel HFCNet, which takes advantage of the heterogeneous features to boost the performance improvement in RSI-SOD. We will elaborate our model in Section III.

III. Proposed Method

In this section, we elaborate the proposed HFCNet. In Section III-A, we give an overview of the proposed HFCNet. In Section III-B, we present the hybrid heterogeneous encoder. In Section III-C, we introduce the design of the AGLI module. Then, we detail the SGAED in Section III-D. Finally, we present our hybrid loss function in Section III-E.

A. Network Overview

As illustrated in Fig. 2, the proposed HFCNet adopts a classic encoder–decoder architecture, which consists of three key components: a hybrid heterogeneous encoder, three AGLI components, and a decoder network with SGAEDs. The network takes optical RSI as input and extracts a set of complementary heterogeneous features through the hybrid heterogeneous encoder. In the AGLI module, local information and global information in these heterogeneous features are adaptively aggregated. Finally, the SGAED modules refine the salient object mask from coarse to fine progressively.

B. Hybrid Heterogeneous Encoder

As shown in Fig. 2, the hybrid heterogeneous encoder initiates with a series of VGG16 [42] blocks (i.e., stages 1–5 of VGG16) to capture local features. VGG16 is widely used in the field of image classification. The model abstracts features into a form suitable for classification through the last max-pooling layer and three fully connected layers. However, RSI-SOD is a pixel-level dense prediction task that requires the preservation of more spatial information. Therefore, we remove the last max-pooling layer and three fully connected layers. In addition, to improve operating efficiency, we adjust the channel dimension in stages 4 and 5 of VGG16 to 256. As the network develops into deeper layers, we incorporate blocks from Swin-B [43] (specifically, we adopt stages 2–4 of Swin-B backbone) to facilitate global context comprehension. These Swin-B stages correspond to encoder stages 3–5 within our framework. For the input image $\mathrm{I} \in \mathbb{R}^{3\times 224\times 224}$, the extracted features of each CNN block are denoted as $f_v^i \in \mathbb{R}^{C_i\times H_i\times W_i}$, where $C_{i=\{1,2,3,4,5\}} = \{64, 128, 256, 256, 256\}$, $H_{i=\{1,2,3,4,5\}} = \{224, 112, 56, 28, 14\}$ and $W_{i=\{1,2,3,4,5\}} = \{224, 112, 56, 28, 14\}$. And the output features of each transformer block are denoted as $f_s^i \in \mathbb{R}^{C_i'\times H_i\times W_i}$, where $C'_{i=\{3,4,5\}} = \{256, 512, 1024\}$, $H_{i=\{3,4,5\}} = \{56, 28, 14\}$, and $W_{i=\{3,4,5\}} = \{56, 28, 14\}$.

C. Adaptive Global–Local Integration Module

In the field of RSI-SOD, global information contributes to the comprehension of the overall semantic context, while local information indicates intricate details of salient objects, so they must be coordinated to obtain higher quality saliency maps. However, directly merging the global information obtained from transformer with the local information extracted by CNN could result in mutual interference between them. To mitigate this interference and effectively merge complementary information, we propose the AGLI module. As shown in Fig. 3, the AGLI module consists of globalwise feature alignment module (GFAM) and pixelwise feature alignment module (PFAM), which are devised to reduce interfering information between heterogeneous features at global and local levels, respectively.

1) Globalwise Feature Alignment Module: The GFAM submodule is designed to align heterogeneous features from a global semantic perspective. As shown in Fig. 3 (left), the output feature map of VGG16 blocks $f_v^i \in \mathbb{R}^{C_i\times H_i\times W_i}$ and the output feature map of Swin-B blocks $f_s^i \in \mathbb{R}^{C_i'\times H_i\times W_i}$ are first fed into GFAM. To achieve a highly summarized description

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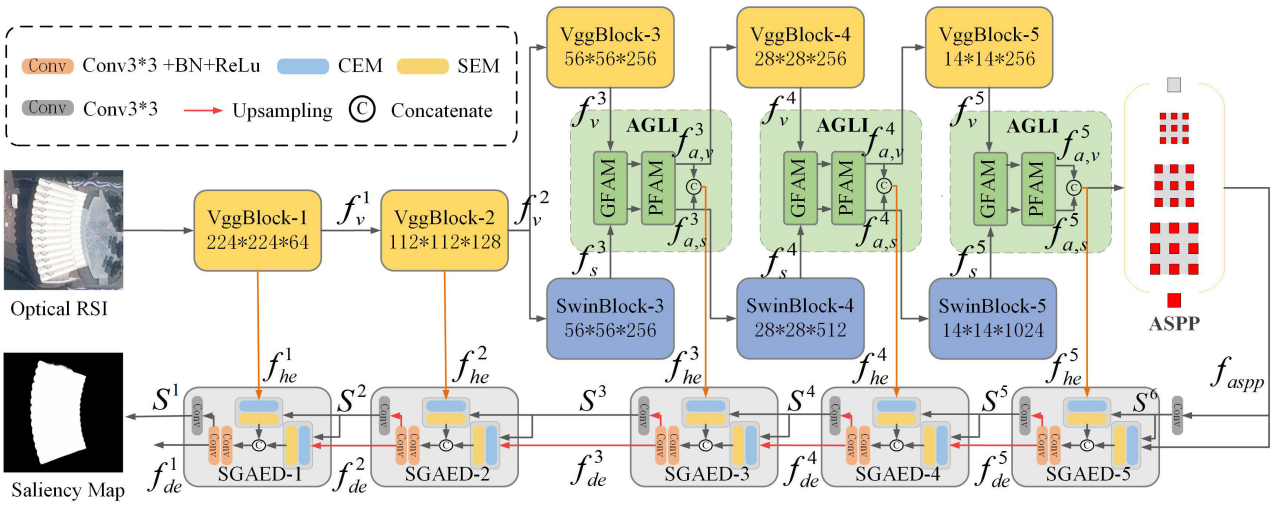


Fig. 2. Overall architecture of the proposed HFCNet. The network consists of three key components: the hybrid heterogeneous encoder, AGLI, and SGAED.

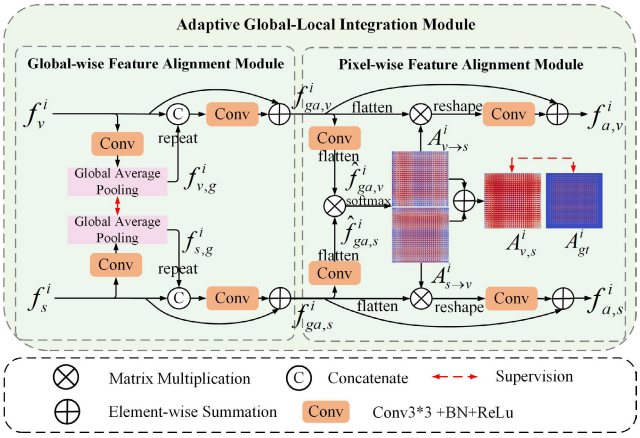


Fig. 3. Internal structure of AGLI.

of global content, we use global average pooling operation to these feature maps, compressing the global distribution information into a single-pixel granular representation. It is formulated as follows:

$$f^i_{v,g} = \mathrm{GAP}_c\big(\mathrm{Conv}\big(f^i_v\big)\big) \tag{1}$$

$$f^i_{s,g} = \mathrm{GAP}_c\big(\mathrm{Conv}\big(f^i_s\big)\big) \tag{2}$$

where $\mathrm{Conv}(\cdot)$ represents a basic convolution block, comprising 3×3 convolution, batch normalization (BN), and ReLu activation function, which is used to align channel dimensions. $\mathrm{GAP}_c(\cdot)$ denotes channelwise global average pooling operation. The $f^i_{v,g} \in \mathbb{R}^{C\times 1\times 1}$ and $f^i_{s,g} \in \mathbb{R}^{C\times 1\times 1}$ represent the global feature representations extracted by CNN and transformer, respectively.

Despite the distinct explicit representations of the two heterogeneous features, their overall tone is consistent. We exploit the latent consistency to constrain heterogeneous features to exclude irrelevant features and define the global-level loss $\mathcal{L}_g$ as follows:

$$\mathcal{L}_g = \sum_{i=3}^{5} \mathcal{L}_{\mathrm{MSE}}\big(f^i_{v,g}, f^i_{s,g}\big) \tag{3}$$

where $\mathcal{L}_{\mathrm{MSE}}(\cdot)$ represents mean square error loss function. Then, we repeat $f^i_{v,g}$ and $f^i_{s,g}$, expanding their dimensions to $C \times H_i \times W_i$. And the global features $f^i_{v,g}$ and $f^i_{s,g}$ are adopted to provide guidance and calibration to their original features, generating globally aligned features $f^i_{\mathrm{ga},v} \in \mathbb{R}^{C_i'\times H_i\times W_i}$ and $f^i_{\mathrm{ga},s} \in \mathbb{R}^{C_i'\times H_i\times W_i}$. It can be formulated as follows:

$$f^i_{\mathrm{ga},v} = \mathrm{Conv}\big(\mathrm{Cat}\big(f^i_{v,g}, f^i_v\big)\big) + f^i_v \tag{4}$$

$$f^i_{\mathrm{ga},s} = \mathrm{Conv}\big(\mathrm{Cat}\big(f^i_{s,g}, f^i_s\big)\big) + f^i_s \tag{5}$$

where $\mathrm{Cat}(\cdot)$ means the concatenation operation. $+$ is the elementwise summation.

2) Pixelwise Feature Alignment Module : The PFAM submodule is designed to align heterogeneous features from a local detail perspective. We expect to use the semantic correlation between the pixel-level representation of heterogeneous features to align them, thereby filtering out noisy information. As shown in Fig. 3 (right), we first apply a convolution operation to $f^i_{\mathrm{ga},v}$ and $f^i_{\mathrm{ga},s}$, and then flatten them to obtain $\hat{f}^i_{\mathrm{ga},v} \in \mathbb{R}^{C\times H_iW_i}$ and $\hat{f}^i_{\mathrm{ga},s} \in \mathbb{R}^{C\times H_iW_i}$, respectively. Subsequently, We calculate the affinity matrix $A^i_{v,s}$ to analyze the pixel correlations between $\hat{f}^i_{\mathrm{ga},v}$ and $\hat{f}^i_{\mathrm{ga},s}$, which is formulated as follows:

$$A^i_{v\to s} = \mathrm{softmax}\big(\hat{f}^{i^T}_{\mathrm{ga},v} \otimes \hat{f}^i_{\mathrm{ga},s}\big) \tag{6}$$

$$A^i_{s\to v} = \mathrm{softmax}\big(\hat{f}^{i^T}_{\mathrm{ga},s} \otimes \hat{f}^i_{\mathrm{ga},v}\big) \tag{7}$$

$$A^i_{v,s} = A^i_{v\to s} + A^i_{s\to v} \tag{8}$$

where $\otimes$ represents matrix multiplication.

We believe the pixel-level correlation in ground truth reflects that found in the heterogeneous features pixels. To determine the affinity matrix between ground-truth pixels, we initially downsample and flatten the ground truth G with size $H \times W$, denoted as $\hat{G}^i \in \mathbb{R}^{1\times H_iW_i}$. Then, the matrix multiplication is applied on $\hat{G}^i$ to obtain the affinity matrix A^i_{gt}

$$A^i_{\mathrm{gt}} = \hat{G}^{iT} \otimes \hat{G}^i. \tag{9}$$

Subsequently, we align A^i_{gt} and $A^i_{v,s}$ to mitigate the noise or interference arising from feature heterogeneity and define the pixel-level loss $\mathcal{L}_p$ as follows:

$$\mathcal{L}_p = \sum_{i=3}^{5} \mathcal{L}_{\mathrm{BCE}}\big(A^i_{\mathrm{gt}}, A^i_{v,s}\big) \tag{10}$$

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where $\mathcal{L}_{\mathrm{BCE}}(\cdot)$ represents binary cross-entropy (BCE) loss function. Ultimately, the correlation matrix is deployed to refine the visual features, i.e.,

$$f_{a,v}^{i} = \mathrm{Conv}\big(\mathrm{reshape}\big(\mathrm{flatten}\big(f_{\mathrm{ga},v}^{i}\big) \otimes A_{v\to s}^{i}\big)\big) + f_{\mathrm{ga},v}^{i} \quad (11)$$

$$f_{a,s}^{i} = \mathrm{Conv}\big(\mathrm{reshape}\big(\mathrm{flatten}\big(f_{\mathrm{ga},s}^{i}\big) \otimes A_{s\to v}^{i}\big)\big) + f_{\mathrm{ga},s}^{i} \quad (12)$$

where flatten$(\cdot)$ represents the operation of flattening a 2-D matrix into a 1-D vector, and reshape$(\cdot)$ denotes the operation of reshaping a 1-D vector to a 2-D matrix. $f_{a,v}^{i}$ and $f_{a,s}^{i}$ represent the globally and locally aligned features from VGG16 and Swin-B blocks, respectively. These features are then fed into the next block of the encoder for continued feature extraction. Simultaneously, we merge them to get the final output f_{he}^{i}, which serves as the input for the decoder. It is formulated as follows:

$$f_{\mathrm{he}}^{i} = \mathrm{Conv}\big(\mathrm{Cat}\big(f_{a,v}^{i}, f_{a,s}^{i}\big)\big). \quad (13)$$

For the output $f_{\mathrm{he}}^{5} \in \mathbb{R}^{c\times h\times w}$ from the final AGLI module, we feed it into the ASPP [44] module to learn multiscale information to further locate salient objects. The final output is represented as $f_{\mathrm{aspp}} \in \mathbb{R}^{c\times h\times w}$, formulated as follows:

$$f_{\mathrm{aspp}} = \mathrm{ASPP}\big(f_{\mathrm{he}}^{5}\big). \quad (14)$$

D. Saliency-Guided Attention Enhanced Decoder

The SGAED module is devised to suppress cluttered background and highlight salient objects, where deep salient cues are used to gradually refine shallow attention maps in both the channel and spatial dimensions. As shown in Fig. 4, SGAED consists of channel enhancement module (CEM) and spatial enhancement module (SEM). In CEM, these salient cues are strategically used to enhance features along the channel dimension, while in SEM they play a crucial role in highlighting the key pixels within the spatial domain.

As illustrated in Fig. 4, the input of the ith SGAED includes the sideout feature map f_{he}^{i} from the ith hybrid heterogeneous encoder, the decoder feature f_{de}^{i+1}, and the salient prediction S^{i+1} from the $(i+1)$th SGAED. To enhance the representation of features, we first input f_{he}^{i} into CEM, yielding a channel enhanced feature. Specifically, we first flatten f_{he}^{i} to $\hat{f}_{\mathrm{he}}^{i} \in \mathbb{R}^{C_i\times H_i W_i}$ and S^{i+1} to $\hat{S}^{i+1} \in \mathbb{R}^{1\times H_i W_i}$. Then, we perform a channelwise correlation analysis between them to obtain the saliency-guided channel importance vector $f_{\mathrm{sg}}^{c} \in \mathbb{R}^{C_i\times 1}$, which reflects the significance of different channels. It is formulated as follows:

$$f_{\mathrm{sg}}^{c} = \hat{f}_{\mathrm{he}}^{i} \otimes \hat{S}^{i+1^{T}}. \quad (15)$$

Subsequently, the original plain channel attention information $f_{\max}^{c} \in \mathbb{R}^{C_i\times 1}$ of feature f_{he}^{i} is generated using max-pooling operation. After applying the multilayer perceptron (MLP) to both f_{sg}^{c} and $f_{\max}^{c}$, we merge them to generate a set of enhanced channel weights CA^{i}. These weights can be used to emphasize important channels while suppressing irrelevant ones. It is formulated as follows:

$$f_{\max}^{c} = \mathrm{GMP}_{c}\big(\hat{f}_{\mathrm{he}}^{i}\big) \quad (16)$$

$$\mathrm{CA}^{i} = \sigma\big(\mathrm{MLP}\big(f_{\mathrm{sg}}^{c}\big) + \mathrm{MLP}\big(f_{\max}^{c}\big)\big) \quad (17)$$

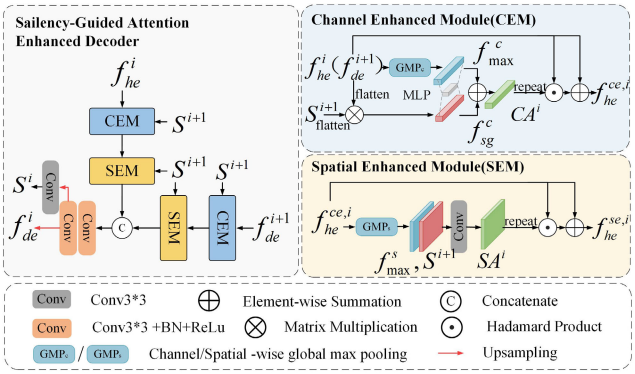


Fig. 4. Internal structure of SGAED.

where $\mathrm{GMP}_{c}(\cdot)$ represents channelwise global max-pooling. $\mathrm{MLP}(\cdot)$ represents an MLP. $\sigma(\cdot)$ denotes the sigmoid function. Finally, we expand the dimensions of CA^{i} to $C_i \times H_i \times W_i$ and use it to selectively filter and refine the original visual information, which is formulated as follows:

$$f_{\mathrm{he}}^{\mathrm{ce},i} = \mathrm{CA}^{i} \odot f_{\mathrm{he}}^{i} + f_{\mathrm{he}}^{i} \quad (18)$$

where $\odot$ denotes the Hadamard product. $f_{\mathrm{he}}^{\mathrm{ce},i}$ represents the channel enhanced feature.

We then input the output $f_{\mathrm{he}}^{\mathrm{ce},i}$ into SEM for further refinement. A set of enhanced spatial weights $\mathrm{SA}^{i} \in \mathbb{R}^{1\times H_i\times W_i}$ are generated guided by the salient cue S^{i+1}, which is formulated as follows:

$$f_{\max}^{s} = \mathrm{GMP}_{s}\big(f_{\mathrm{he}}^{\mathrm{ce},i}\big) \quad (19)$$

$$\mathrm{SA}^{i} = \sigma\big(\mathrm{Conv}\big(\mathrm{Cat}\big(f_{\max}^{s}, S^{i+1}\big)\big)\big) \quad (20)$$

where $\mathrm{GMP}_{s}(\cdot)$ represents spatialwise global max pooling. We repeat SA^{i} to match the dimensions $C_i \times H_i \times W_i$. Then, we use it to filter the feature map along the spatial dimension, emphasizing important pixels and generating the spatial enhanced feature $f_{\mathrm{he}}^{\mathrm{se},i}$. It is formulated as follows:

$$f_{\mathrm{he}}^{\mathrm{se},i} = \mathrm{SA}^{i} \odot f_{\mathrm{he}}^{\mathrm{ce},i} + f_{\mathrm{he}}^{\mathrm{ce},i}. \quad (21)$$

The same operation is also applied to the output feature f_{de}^{i+1} generated from the $(i+1)$th SGAED, enhancing the decoder features $f_{\mathrm{de}}^{\mathrm{se},i+1}$. Finally, the ith decoder output f_{de}^{i} is generated by merging the output $f_{\mathrm{he}}^{\mathrm{se},i}$ and $f_{\mathrm{de}}^{\mathrm{se},i+1}$ as follows:

$$f_{\mathrm{de}}^{i} = \begin{cases} f_{\mathrm{aspp}}, & \text{if } i = 6 \\ \mathrm{Up}_{2}\big(\mathrm{Conv}\big(\mathrm{Cat}\big(f_{\mathrm{he}}^{\mathrm{se},i}, f_{\mathrm{de}}^{\mathrm{se},i+1}\big)\big)\big), & \text{if } i \le 5 \end{cases} \quad (22)$$

where $\mathrm{Up}_{2}(\cdot)$ is the $2\times$ upsampling operation. For the output f_{de}^{i} of the ith SGAED, we use a 3×3 convolutional layer to project it into a saliency map S^{i}, and we select S^{1} as the final result. It is formulated as follows:

$$S^{i} = \begin{cases} \sigma\big(\mathrm{Conv}_{3\times 3}\big(f_{\mathrm{aspp}}\big)\big), & \text{if } i = 6 \\ \sigma\big(\mathrm{Conv}_{3\times 3}\big(f_{\mathrm{de}}^{i}\big)\big), & \text{if } i \le 5. \end{cases} \quad (23)$$

E. Loss Function

To accelerate network convergence and enhance feature learning, we construct a comprehensive loss function, including BCE loss, intersection-over-union (IoU) loss, global-level loss, and pixel-level loss.

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1) Saliency Supervision: We adopt the BCE loss and IoU loss to supervise all the generated intermediate saliency maps, which is formulated as follows:

$$\mathcal{L} = \frac{1}{6}\sum_{i=1}^{6}(\mathcal{L}_{\mathrm{BCE}}(\mathrm{Up}_{k_i}(S^i), G) + \mathcal{L}_{\mathrm{IoU}}(\mathrm{Up}_{k_i}(S^i), G)) \quad (24)$$

where G refers to the ground-truth map, and $\mathrm{Up}_{k_i}(\cdot)$ denotes upsampling operation by a factor of k_i, defined as

$$k_i = \begin{cases} 2^4, & \text{if } i = 6 \\ 2^{i-1}, & \text{if } i \leq 5. \end{cases}$$

2) Heterogeneous Features Supervision: To further enhance the model's capability in aligning heterogeneous features, we specifically incorporate a global-level loss $\mathcal{L}_g$ and a pixel-level loss $\mathcal{L}_p$ into our loss function, where $\mathcal{L}_g$ and $\mathcal{L}_p$ are detailed in (3) and (10).

Finally, the overall loss function $\mathcal{L}_{\mathrm{total}}$ is formulated as follows:

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L} + \mathcal{L}_g + \mathcal{L}_p. \quad (25)$$

IV. Experiment

In this section, we first introduce the experiment settings in Section IV-A. Subsequently, we engage in an extensive comparison of the proposed model with the state-of-the-art saliency models in Section IV-B. Then, the ablation experiments are discussed in Section IV-C. Furthermore, we analyze the complexity of the proposed model and modules in Section IV-D. In Section IV-E, we investigate the effect of the seed value selection on performance. Finally, failure cases and analysis are presented in Section IV-F.

A. Experimental Settings

1) Datasets: The experiments are conducted on three public datasets, including ORSSD [37], EORSSD [17], and ORSI4199 [38].

ORSSD comprises 800 samples with the corresponding pixelwise ground truths, of which 600 are used for training and 200 for testing.

EORSSD is an expanded version of ORSSD, containing 1400 training samples and 600 testing samples. Compared with ORSSD, this dataset features a more diverse and challenging set of samples.

ORSI4199, the latest released dataset in the RSI-SOD domain, contains 2000 training samples and 2199 testing samples. Renowned for its extensive scale and the complexity of its scenarios, ORSI4199 is considered the most challenging dataset in the field of RSI-SOD.

2) Implementation Details: We implement our HFCNet by PyTorch with one NVIDIA RTX 3090 Ti GPU. For data augmentation, we follow the approach [37] to flip and rotate each image in the dataset, creating seven additional variations in the original training data. This technique effectively increases the size of training set by eight times. During training, input images are resized to 224×224. The encoder network of HFCNet is initialized using pretrained parameters from VGG16 and Swin-B models. Besides, we use the Adam optimizer [45] with a batch size of 8, an initial learning rate of 7×10^{-5}, and train for 40 epochs.

3) Evaluation Metrics: We select widely recognized evaluation metrics to validate the efficiency of our model.

1) *Precision and Recall Curve [19]:* Using thresholds ranging from 0 to 255 to map the saliency map into binary masks, we accurately calculate the precision and recall at different thresholds and plot the PR curve. The combinations of precision (P) and recall (R) at various thresholds are defined as follows:
$$P = \frac{|S \cap G|}{|S|}, \quad R = \frac{|S \cap G|}{|G|} \quad (26)$$
where S and G are the binary saliency mask at different thresholds and ground truth, respectively.
2) *S-Measure [50]:* S_α comprehensively evaluates the structural similarity at object-aware (S_o) and region-aware level (S_r), which is defined as follows:
$$S = \alpha \times S_o + (1 - \alpha) \times S_r \quad (27)$$
where $\alpha = 0.5$ is used to balance these two similarities.
3) *F-Measure [19]:* F_β represents the weighted average of precision and recall. F_β is defined as follows:
$$F_\beta = \frac{(1 + \beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R} \quad (28)$$
where β^2 is set to 0.3, underscoring an emphasis on precision.
4) *E-Measure [51]:* E_ξ simultaneously takes into account both the local pixel-level match information and the global image-level statistics. E_ξ is defined as follows:
$$E_\xi = \frac{1}{H \times W}\sum_{i=1}^{H \times W} \xi_S(i) \quad (29)$$
where $\xi_S(i)$ describes the relationship between saliency prediction and ground truth after removing the means. H and W denote the image height and width, respectively.
5) *MAE [52]:* Mean absolute error quantifies the average error between the predicted and actual values for each pixel within an image. MAE is defined as follows:
$$\mathrm{MAE} = \frac{1}{H \times W}\sum_{i=1}^{H \times W} |S_i - G_i| \quad (30)$$
where S_i and G_i represent the value at the ith pixel in the predicted saliency map and ground truth, respectively. The score of MAE is smaller, and the performance is better.

B. Comparison With State-of-the-Art Methods

1) Comparison Methods: We conduct a comprehensive comparison of our method with 18 state-of-the-art NSI-SOD and RSI-SOD methods on three challenging datasets. As presented in Table I, these compared methods include two traditional NSI-SOD methods (RRWR [1] and RCRR [2]), three CNN-based NSI-SOD methods (EGNet [41], MINet [4], and GatedNet [5]), two hybrid-based RSI-SOD methods

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TABLE I

QUANTITATIVE COMPARISONS OF OUR METHOD WITH 18 STATE-OF-THE-ART METHODS, INCLUDING TWO TRADITIONAL NSI-SOD METHODS (T-NSI), THREE CNN-BASED NSI-SOD METHODS (C-NSI), TWO HYBRID-BASED RSI-SOD METHOD (H-RSI), ONE TRANSFORMER-BASED RSI-SOD METHOD (T-RSI), AND TEN CNN-BASED RSI-SOD METHODS (C-RSI). THE SUFFIXES "-V" AND "-R" INDICATE THAT THE BACKBONE OF THE MODEL IS VGG [42] OR RESNET [46]. THE TOP THREE RESULTS ARE IN RED, BLUE, AND GREEN, RESPECTIVELY. "-" MEANS NO DATA AVAILABLE

Method	Publication	Type	FPS	ORSSD				EORSSD				ORSI4199			
				$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$MAE \downarrow$	$S_\alpha \uparrow$	$F_\beta \uparrow$	$E_\xi \uparrow$	$MAE \downarrow$
RRWR [1]	2015 CVPR	T-NSI	0.3	0.6835	0.5590	0.7649	0.1324	0.5992	0.3993	0.6894	0.1677	0.6416	0.5407	0.7116	0.1717
RCRR [2]	2018 TIP	T-NSI	0.3	0.6849	0.5591	0.7651	0.1277	0.6007	0.3995	0.6882	0.1644	0.6491	0.5480	0.7192	0.1637
EGNet [3]	2019 ICCV	C-NSI	-	0.8721	0.8332	0.9731	0.0216	0.8601	0.7880	0.9570	0.0110	0.8516	0.8371	0.9241	0.0385
MINet [4]	2020 CVPR	C-NSI	12	0.9040	0.8761	0.9545	0.0144	0.9040	0.8344	0.9442	0.0093	0.8116	0.7988	0.8961	0.0504
GatedNet [5]	2020 ECCV	C-NSI	25	0.9186	0.8871	0.9664	0.0137	0.9114	0.8566	0.9610	0.0095	0.8545	0.8450	0.9256	0.0393
LVNet-V [37]	2019 TGRS	C-RSI	1.4	0.8815	0.8263	0.9456	0.0207	0.8630	0.7794	0.9254	0.0146	-	-	-	-
DAFNet-V [17]	2021 TIP	C-RSI	26	0.9191	0.8928	0.9771	0.0113	0.9166	0.8614	0.9861	0.0060	0.8492	0.8348	0.9181	0.0422
MCCNet-V [18]	2021 TGRS	C-RSI	-	0.9437	0.9155	0.9800	0.0087	0.9327	0.8904	0.9755	0.0066	-	-	-	-
MJRBM-V [38]	2022 TGRS	C-RSI	32	0.9204	0.8842	0.9623	0.0163	0.9197	0.8656	0.9646	0.0099	0.8593	0.8493	0.9311	0.0374
EMFINet-V [47]	2022 TGRS	C-RSI	25	0.9366	0.9002	0.9737	0.0109	0.9290	0.8720	0.9711	0.0084	0.8692	0.8608	0.9373	0.0327
CorrNet-V [48]	2022 TGRS	C-RSI	100	0.9380	0.9129	0.9790	0.0098	0.9289	0.8778	0.9696	0.0083	0.8626	0.8560	0.9333	0.0366
ERPNet-V [41]	2023 TCYB	C-RSI	50	0.9254	0.8974	0.9710	0.0135	0.9210	0.8632	0.9603	0.0089	0.8642	0.8551	0.9284	0.0376
ACCoNet-V [15]	2023 TCYB	C-RSI	81	0.9437	0.9149	0.9796	0.0088	0.9290	0.8837	0.9727	0.0074	0.8775	0.8687	0.9412	0.0314
AESINet-V [14]	2023 TGRS	C-RSI	25	0.9460	0.9183	0.9828	0.0086	0.9358	0.8838	0.9751	0.0079	0.8702	0.8670	0.9403	0.0325
ASTTNet [40]	2023 TGRS	T-RSI	13	0.9347	0.9060	0.9794	0.0094	0.9253	0.8741	0.9580	0.0060	0.8827	0.8788	0.9512	0.0273
Ours-V		H-RSI	38	0.9535	0.9256	0.9873	0.0071	0.9445	0.8955	0.9835	0.0052	0.8830	0.8829	0.9520	0.0279
MJRBM-R [38]	2022 TGRS	C-RSI	22	0.9211	0.8885	0.9686	0.0145	0.9091	0.8555	0.9655	0.0099	0.8582	0.8511	0.9343	0.0372
RRNet-R [39]	2022 TGRS	C-RSI	109	0.9339	0.9011	0.9722	0.0113	0.9266	0.8743	0.9665	0.0082	0.8585	0.8500	0.9286	0.0367
HFANet-R [16]	2022 TGRS	H-RSI	26	0.9399	0.9117	0.9770	0.0092	0.9380	0.8876	0.9740	0.0071	0.8767	0.8700	0.9431	0.0314
EMFINet-R [47]	2022 TGRS	C-RSI	25	0.9432	0.9155	0.9813	0.0095	0.9319	0.8742	0.9712	0.0075	0.8712	0.8636	0.9403	0.0313
ERPNet-R [41]	2023 TCYB	C-RSI	50	0.9352	0.9036	0.9738	0.0114	0.9252	0.8743	0.9665	0.0082	0.8636	0.8528	0.9292	0.0388
ACCoNet-R [15]	2023 TCYB	C-RSI	-	0.9428	0.9149	0.9819	0.0087	0.9302	0.8821	0.9759	0.0067	0.8805	0.8688	0.9424	0.0320
AESINet-R [14]	2023 TGRS	C-RSI	23	0.9455	0.9160	0.9814	0.0085	0.9347	0.8792	0.9757	0.0064	0.8755	0.8726	0.9459	0.0305
ADSTNet-R [49]	2024 JSTARS	H-RSI	-	0.9379	0.9124	0.9807	0.0086	0.9311	0.8804	0.9769	0.0065	0.8710	0.8698	0.9433	0.0318
Ours-R		H-RSI	49	0.9521	0.9247	0.9885	0.0073	0.9407	0.8864	0.9793	0.0054	0.8838	0.8833	0.9539	0.0277

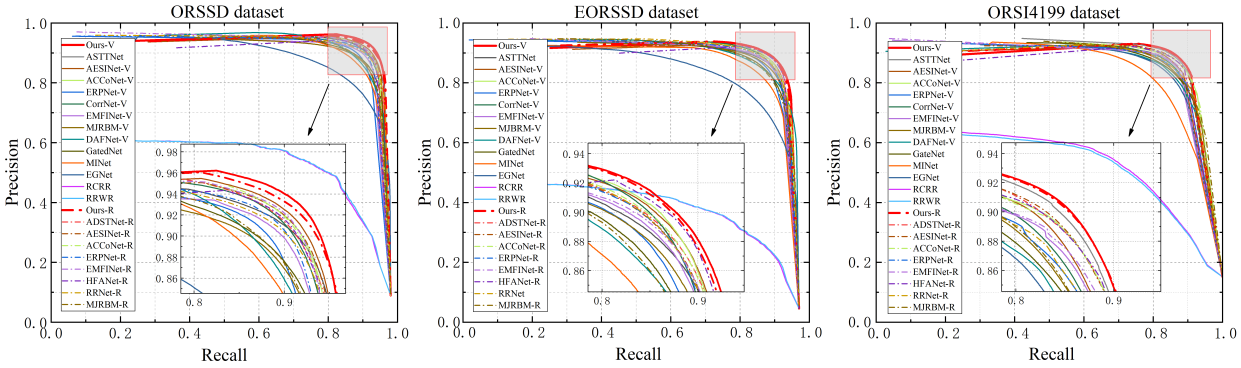


Fig. 5. PR curves' comparison of different models on three benchmark datasets. The solid lines represent T-NSI methods, VGG-based C-RSI methods (-V), and T-RSI methods, while the dotted lines denote ResNet-based C-RSI and H-RSI methods (-R).

(HFANet [16] and ADSTNet [49]), one transformer-based RSI-SOD method (ASTTNet [40]), and ten CNN-based RSI-SOD methods (LVNet [37], DAFNet [17], MCCNet [18], MJRBM [38], EMFINet [47], CorrNet [48], RRNet [39], ERPNet [41], ACCoNet [15], and AESINet [14]). For a fair comparison, all the saliency maps are collected from publicly available results provided by the authors or computed by running the source code released by the authors.

2) Quantitative Comparison: Table I presents the quantitative comparison results of our method with 18 other state-of-the-art methods on three datasets, from which we can see that the proposed method outperforms these advanced methods. Among the 12 metrics listed in Table I, Ours-V achieves ten best performances and two second-best performances, while Ours-R achieves 11 best performances and one second-best performance. Although some results are slightly lower than other methods, the performance gaps are not significant. When compared with the latest transformer-based method (ASTTNet [40]), Ours-V shows improvements of around 1.88% in S_α, 1.96% in F_β, 0.79% in E_ξ, and 0.0023 in MAE on the ORSSD dataset. In comparison to the latest hybrid-based method (ADSTNet-R [49]), our method (Ours-R) demonstrates an improvement of 1.42% in S_α, 1.23% in F_β, 0.78% in E_ξ, and 0.0013 in MAE on the ORSSD dataset. Similar trends are observed across other datasets. From the data presented, it is evident that our method remains competitive

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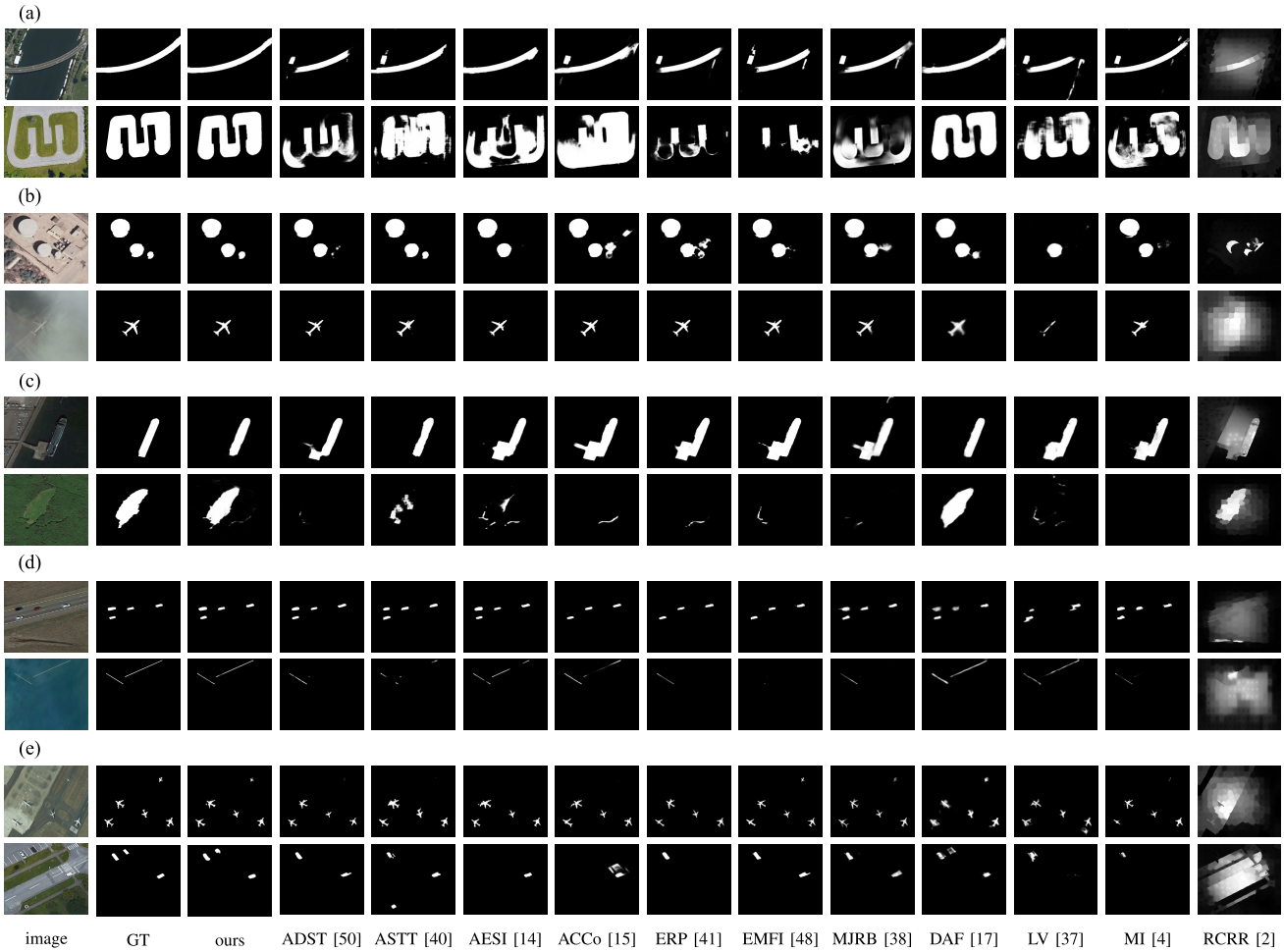


Fig. 6. Visual comparison of our method (Ours-V) and several representative saliency models (-V) on EORSSD testing data. (a) Objects with irregular topology. (b) Objects with interference. (c) Objects with complex backgrounds. (d) Tiny objects. (e) Multiple objects.

when compared with these latest methods. In Table I, we also compare our method with these state-of-the-art methods in terms of FPS. Ours-V and Ours-R achieve processing speeds of 38 and 49 frames/s, respectively. Overall, our method has demonstrated significant detection accuracy and rapid processing speed. In addition, we present a comparison of the PR curves of our method with several state-of-the-art methods in Fig. 5. The PR curve of our model is closest to the coordinate (1,1), which indicates that our model achieves a higher precision while also maintaining a higher recall.

3) Qualitative Comparison: We show several visual representative examples of optical RSI in Fig. 6. It can be observed that our model exhibits more complete and accurate detection of salient objects in complex scenes (CSs). As illustrated in Fig. 6(a), salient objects often have irregular geometric shapes with varying widths at different positions, and they may span across the entire image. This presents a significant challenge for detecting salient objects. Compared with other methods, the proposed method can detect salient objects more completely, which is attributed to its ability to model long-range relationships effectively. Fig. 6(b) displays objects with interference, and (c) shows objects with complex backgrounds. In these challenging scenarios, our approach efficiently reduces interference from background noise, while other methods mistakenly highlight background regions. This mainly relies on the proposed SGAED module, which offers valuable location cues of salient objects to shallow features, guiding the decoder to focus more on important regions. Fig. 6(d) showcases tiny objects. While the existing methods face difficulties in accurately locating and detailing these objects, our method is well-equipped to address this challenge. In the proposed model, the transformer branch effectively captures global information to locate target objects accurately, and the CNN branch focuses on intricate local information

TABLE II

ATTRIBUTES-BASED PERFORMANCE ASSESSMENT ON THE ORSI4199 DATASET. THE TOP THREE F_β ARE HIGHLIGHTED IN RED, BLUE, AND GREEN, RESPECTIVELY. FOR DETAILED DEFINITIONS OF EACH ATTRIBUTE, REFER TO [38]

Att.	MJRBM [38]	RR [39]	Corr [48]	ERP [41]	ACCo [15]	AESI [14]	ASTT [40]	ADST [49]	ours
BSO	0.9166	0.8811	0.8891	0.8946	0.9091	0.9109	0.9180	0.9180	0.9303
CS	0.8961	0.8702	0.8835	0.8800	0.8974	0.9068	0.9090	0.9012	0.9163
CSO	0.8899	0.8649	0.8728	0.8727	0.8931	0.9001	0.8900	0.8940	0.9008
ISO	0.8937	0.8594	0.8742	0.8837	0.8997	0.9093	0.9230	0.8995	0.9232
LCO	0.7916	0.7710	0.7852	0.7925	0.8048	0.8136	0.7850	0.8092	0.8220
MSO	0.8585	0.8288	0.8528	0.8468	0.8611	0.8579	0.8570	0.8567	0.8628
NSO	0.8748	0.8392	0.8724	0.8467	0.8814	0.8922	0.8980	0.8895	0.9098
OC	0.8422	0.8042	0.8409	0.8378	0.8337	0.8349	0.8370	0.8503	0.8474
SSO	0.8091	0.7682	0.8051	0.8015	0.8065	0.8079	0.8000	0.8056	0.8156
Avg	0.8636	0.8319	0.8529	0.8507	0.8652	0.8704	0.8690	0.8693	0.8809

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TABLE III

ABLATION EXPERIMENT RESULTS BETWEEN DIFFERENT MODULES ON THE ORSSD AND EORSSD DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD

No.	CNN	TF	AGLI	SGAED	ORSSD			EORSSD		
					S_α	F_β	E_ξ	S_α	F_β	E_ξ
1	✓				0.9299	0.8991	0.9701	0.9229	0.8607	0.9569
2		✓			0.9326	0.8966	0.9749	0.9245	0.8626	0.9646
3	✓	✓			0.9443	0.9155	0.9765	0.9249	0.8791	0.9702
4	✓	✓	✓		0.9509	0.9187	0.9823	0.9411	0.8923	0.9804
5	✓	✓		✓	0.9472	0.9184	0.9797	0.9404	0.8901	0.9811
6	✓	✓	✓	✓	**0.9535**	**0.9256**	**0.9873**	**0.9445**	**0.8955**	**0.9835**

to improve the details of objects. Collaboration between the two branches ensures completeness and accuracy of detection. Fig. 6(e) presents multiple objects. It is notable that many advanced methods suffer from the problem of missing targets. In contrast, our method can detect all the objects by establishing semantic relationships between different locations within the image, which helps associate objects of the same category.

4) Attribute-Based Performance on ORSI-4199: In the ORSI4199 dataset, each image has challenge attributes that reflect typical challenges in optical RSI. Table II presents the performance scores of the proposed model in comparison to several most state-of-the-art models on the F_β metric. The results show that our model ranks first in eight out of nine attributes. Besides, our model achieves the highest overall average performance, which indicates that the proposed model outperforms the existing models in most challenging situations. Particularly, our model outperforms other models when dealing with complex salient objects (CSOs) and narrow salient objects (NSOs), demonstrating its strong capability in modeling irregular topology. Our model also shows its superiority in CSs, showcasing its ability to recognize salient objects from cluttered backgrounds.

C. Ablation Experiments

In this section, we conducted comprehensive ablation experiments on the ORSSD and EORSSD datasets to validate the effectiveness of the proposed modules. Specifically, our investigations include ablation experiments between different modules, ablation experiments within individual modules, and ablation experiments on the comprehensive loss function.

1) Ablation Experiments Between Different Modules: To verify the effectiveness of four components in our model (i.e., CNN branch, transformer branch, AGLI module, and SGAED module), we designed six variants: 1) CNN (using only the CNN branch as the encoder, with basic convolutional blocks replacing the decoder); 2) TF (using only the transformer branch as the encoder, with basic convolutional blocks replacing the decoder); 3) CNN + TF (using the hybrid heterogeneous encoder combining CNN and transformer, with the features from the two branches directly concatenated to enable interaction); 4) CNN + TF + AGLI ("CNN + TF" combined with the AGLI module); 5) CNN + TF + SGAED ("CNN + TF" combined with the SGAED module); and 6) CNN + TF + AGLI + SGAED (complete HFCNet). We report the quantitative results in Table III. On the EORSSD dataset, we observe that "CNN" only achieves 0.9229 on S_α, 0.8607 on F_β, and 0.9569 on E_ξ. In comparison, "TF" records scores of 0.9245, 0.8626, and 0.9646 on these metrics. When combining CNN and transformer (i.e. "CNN + TF"), the performance shows a slight improvement. The proposed AGLI module (i.e., "CNN + TF + AGLI") increases the "CNN + TF" by 1.62%, 1.32%, and 1.02% on these three metrics, while the deployment of the SGAED module (i.e., "CNN + TF + SGAED") improves "CNN + TF" by 1.55%, 1.1%, and 1.09% on these three metrics. With the joint cooperation of AGLI and SGAED, the complete HFCNet significantly outperforms "CNN + TF," achieving improvements of 1.96%, 1.64%, and 1.33% on these three metrics, respectively. Similar trends are observed on the ORSSD dataset. It is evident that the overall performance of the model continues to improve with the addition of each module, which highlights the effectiveness of the proposed modules.

In addition, the visual examples illustrated in Fig. 7(a) offer an intuitive demonstration of the superiority of the proposed modules. In the first row of images, the branch structure of the river is complex. Variants lacking AGLI present an obvious absence in the slender branch of the river, while our model can detect the object more completely, which shows that

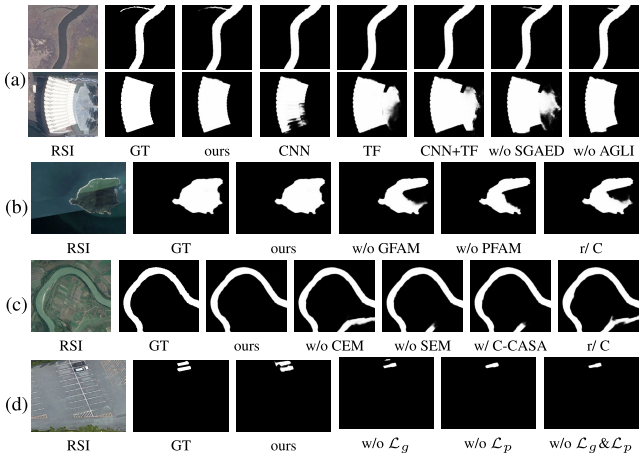


Fig. 7. Visual examples of ablation experiments on the EORSSD dataset. (a) Between different modules. (b) Inside AGLI module. (c) Inside SGAED module. (d) Between different loss functions.

TABLE IV

RESULTS OF ABLATION EXPERIMENTS WITHIN THE AGLI AND SGAED MODULES. THE BEST RESULTS ARE IN BOLD. (a) COMPONENTS IN AGLI. (b) COMPONENTS IN SGAED

Model variants	ORSSD			EORSSD		
	S_α	F_β	E_ξ	S_α	F_β	E_ξ
(a)						
ours	**0.9535**	**0.9256**	**0.9873**	**0.9445**	**0.8955**	**0.9835**
w/o GFAM	0.9454	0.9169	0.9798	0.9403	0.8882	0.9798
w/o PFAM	0.9506	0.9223	0.9833	0.9423	0.8920	0.9802
r/ C	0.9440	0.9155	0.9765	0.9249	0.8791	0.9702
(b)						
ours	**0.9535**	**0.9256**	**0.9873**	**0.9445**	**0.8955**	**0.9835**
w/o CEM	0.9490	0.9192	0.9816	0.9422	0.8884	0.9766
w/o SEM	0.9470	0.9195	0.9833	0.9391	0.8909	0.9809
w/ C-CASA	0.9472	0.9194	0.9814	0.9405	0.8891	0.9774
r/ C	0.9422	0.9135	0.9736	0.9307	0.8814	0.9679

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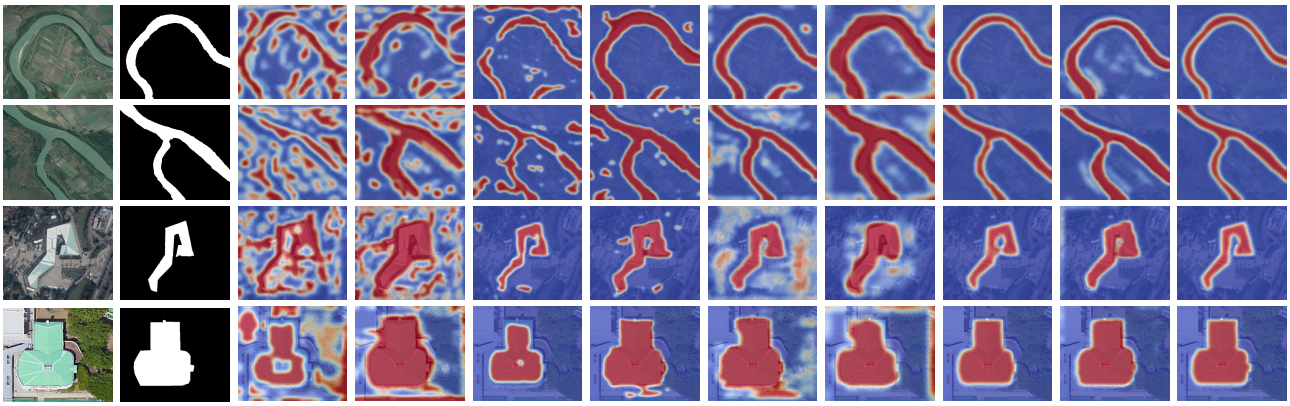


Fig. 8. Visual comparison of feature maps from different modules. (a) Optical RSI. (b) Ground truth. (c) Features f_v^4 from the CNN branch. (d) Enhanced CNN features $f_{a,v}^4$. (e) Features f_s^4 from the transformer branch. (f) Enhanced transformer features $f_{a,s}^4$. (g) Fused features f_{he}^4 from the AGLI module. (h) Output features f_{de}^5 from fifth SGAED. (i) Saliency map S^5. (j) Output features f_{de}^4 from fourth SGAED. (k) Saliency map S^4.

the AGLI module is crucial to improve the model's ability to handle irregular structures. In the second row of images, salient objects are accompanied by a cluttered background. For the variant without SGAED, it can be observed that the results contain a large number of background, but our model can accurately identify salient objects from cluttered backgrounds. It proves the importance of SGAED in eliminating interference.

2) Ablation Experiment Inside the AGLI Module: To verify the effectiveness of key components in AGLI, we provide another three variants of AGLI: 1) w/o GFAM (AGLI without GFAM); 2) w/o PFAM (AGLI without PFAM); and 3) r/ C (replace AGLI with the operation of convolution after concatenation). The experimental results are reported in Table IV(a). It can be found that removing either GFAM or PFAM leads to a decline in performance, which highlights the necessity of aligning both the global information and local information. Aligning global and local information helps eliminate mutual interference among heterogeneous features, enabling the effective integration of local and global information. It is worth noting that when the simple direct concatenation method is used instead of the AGLI module, there is a drop in performance. This indicates that there are significant differences between the two heterogeneous features, making direct and effective fusion challenging. Therefore, we conclude that the essence of the AGLI module lies in its ability to gradually narrow the semantic gap between two heterogeneous features from both a global semantic and a local detail perspective, thereby facilitating a more effective fusion. The visual effect depicted in Fig. 7(b) further confirms the effectiveness of this module. Models using three different versions of the AGLI module perform poorly in detecting irregularly shaped objects, while our method (ours) can accurately detect them.

Fig. 8 shows a visual comparison of features pre- and postprocessing by the fourth AGLI module. Fig. 8(c) and (e) displays features extracted by CNN (i.e., f_v^4) and transformer (i.e., f_s^4), respectively. The CNN features f_v^4 focus on local information, while the transformer features f_s^4 emphasize global information. Through the processing of the AGLI module, (d) shows that the enhanced CNN features (i.e., $f_{a,v}^4$) not only retain their sensitivity to local information but also integrate global information. Similarly, the enhanced transformer features (i.e., $f_{a,s}^4$) in column (f) maintain their global perspective information while also improving their ability to recognize local details. Furthermore, (g) presents the fused features (i.e., f_{he}^4) from the AGLI module, which combine the local and global information, significantly outperforming the other columns. This also demonstrates that AGLI module can effectively combine the strengths of CNN and transformer, accurately perceiving complex and irregular topology structures.

TABLE V

ABLATION EXPERIMENT RESULTS EVALUATING THE EFFECTIVENESS OF THE COMPREHENSIVE LOSS FUNCTION. THE BEST RESULTS ARE SHOWN IN BOLD

No.	BCE	IoU	$\mathcal{L}_g$	$\mathcal{L}_p$	ORSSD			EORSSD		
					S_α	F_β	E_ξ	S_α	F_β	E_ξ
1	✓	✓			0.9476	0.9175	0.9705	0.9406	0.8914	0.9803
2	✓	✓	✓		0.9509	0.9232	0.9834	0.9439	0.8940	0.9809
3	✓	✓		✓	0.9480	0.9205	0.9811	0.9412	0.8913	0.9782
4	✓	✓	✓	✓	**0.9535**	**0.9256**	**0.9873**	**0.9445**	**0.8955**	**0.9835**

TABLE VI

ABLATION EXPERIMENT RESULTS EXPLORING THE IMPACT OF INPUT IMAGE RESOLUTION ON PERFORMANCE

Resolution	ORSSD			EORSSD			ORSI4199		
	S_α	F_β	E_ξ	S_α	F_β	E_ξ	S_α	F_β	E_ξ
224×224	0.9535	0.9256	0.9873	0.9445	0.8955	0.9835	0.8830	0.8829	0.9520
256×256	0.9543	0.9312	0.9894	0.9452	0.8969	0.9836	0.8857	0.8876	0.9539
288×288	0.9550	0.9325	0.9897	0.9481	0.9023	0.9845	0.8868	0.8908	0.9574

3) Ablation Experiment Inside the SGAED Module: To verify the effectiveness of the internal structure of SGAED, we build another four variants of SGAED: 1) w/o CEM (SGAED without CEM); 2) w/o SEM (SGAED without SEM); 3) w/ C-CASA (SGAED replacing CEM and SEM with conventional channel attention and spatial attention mechanisms); and 4) r/ C (replace SGAED with the operation of convolution after concatenation). The experimental results are reported in Table IV(b). We observe that the removal of either CEM

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or SEM leads to a decline in model performance, thereby confirming the effectiveness of these two modules. When CEM and SEM are replaced with conventional channel attention and spatial attention, there is also a decrease in performance. The results indicate that such structures play a secondary role, but the guidance of deep salient cues on shallow decoders has a decisive impact on improving performance. Besides, replacing the SGAED module with basic convolution blocks also leads to decreased performance, which can be attributed to directly merging deep and shallow features introducing a large amount of redundant information. Therefore, we believe that the core advantage of the SGAED module is that the deep salient cues guide the shallow decoders to focus on important areas and suppress irrelevant areas. As shown in Fig. 7(c), the other four variants all highlight the background incorrectly, while our method (ours) successfully avoids this error. This effectively demonstrates the SGAED module's capability.

To provide further evidence of the SGAED module's effectiveness, a visual comparison of features pre- and post-processing by the fourth SGAED module are displayed in Fig. 8. As shown in Fig. 8(g) displays the shallow features f_{he}^4 from the fourth encoder, characterized by rich details but also a significant amount of redundant background information. Fig. 8(h) and (i) presents the deep features f_{de}^5 and the corresponding saliency map S^5 from the fifth SGAED module, which provide a rough localization of salient objects. Fig. 8(j) and (k) shows the output features f_{de}^4 and saliency maps S^4 from the fourth SGAED module. Notably, under the guidance of the saliency map S^5, the background noise in f_{de}^4 is substantially mitigated while detailed features are enhanced. This underscores the effectiveness of the proposed SGAED in eliminating complex background noise and preserving vital detail when integrating deep and shallow features.

4) Ablation Experiment on the Comprehensive Loss Function: To evaluate the individual contribution of each content in our comprehensive loss function, we offer three variants of loss for network training: 1) $\mathcal{L}$ (BCE loss combined with IoU loss); 2) $\mathcal{L} + \mathcal{L}_g$ ($\mathcal{L}$ combined with the global-level loss); and 3) $\mathcal{L} + \mathcal{L}_p$ ($\mathcal{L}$ combined with the pixel-level loss). The experimental results are shown in Table V. On the EORSSD dataset, we initially trained the model using only BCE loss and IoU loss, achieving a score of 0.9406 on S_α, 0.8914 on F_β, and 0.9803 on E_ξ. The performance of the model is further enhanced with the addition of either $\mathcal{L}_g$ or $\mathcal{L}_p$. By incorporating these four loss for training, our model achieved its optimal performance, reaching 0.9445 on S_α, 0.8955 on F_β, and 0.9835 on E_ξ. The same trend can be observed on the ORSSD dataset. The visual effect shown in Fig. 7(d) also confirms the superiority of our comprehensive loss function.

5) Impact of Input Image Resolution on Performance: We also investigate the influence of the input image resolution on performance. The experimental results are shown in Table VI. It can be seen that as the image size increases, the performance improves as well. This may be attributed to the fact that higher resolution images can provide more detailed information, thereby enhancing performance. However, processing images with high resolution requires much computational resources and time. Therefore, we ultimately conduct experiments with a resolution of 224 × 224 pixels.

TABLE VII

COMPLEXITY ANALYSIS OF THE PROPOSED MODULES AND COMPARATIVE RESULTS OF THE PROPOSED MODELS WITH SEVERAL ADVANCED MODELS. TO MEET MAINSTREAM EVALUATION STANDARDS, ALL MODELS INVOLVED IN THE COMPARISON ARE VGG VERSIONS. (a) COMPLEXITY ANALYSIS OF THE PROPOSED MODULES. (b) COMPARISON OF COMPLEXITY WITH SOME STATE-OF-THE-ART METHODS

Models	FLOPs	Params
(a)		
CNN	13.86 G	10.46 M
TF	68.87 G	86.28 M
CNN+TF	82.73 G	96.74 M
CNN+TF+AGLI	109.07 G(↑ 26.34)	134.95 M(↑ 38.21)
CNN+TF+AGLI+SGAED	120.41 G(↑ 11.34)	140.75 M(↑ 5.80)
(b)		
MCCNet [18]	117.15 G	67.65 M
EMFINet [47]	176.87 G	95.09 M
ERPNet [41]	131.63 G	77.19 M
ACCoNet [15]	184.50 G	102.55 M
AESINet [14]	53.42 G	41.05 M
ASTTNet [40]	43.12 G	23.35 M
ADSTNet [49]	62.09 G	27.72 M
ours	120.41 G	140.75 M

D. Computational Complexity Analysis

We use FLOPs and network parameters as metrics to evaluate the complexity of the proposed model and modules. As shown in Table VII(a), the foundational architecture that combines the CNN branch and transformer branch (i.e., "CNN + TF"), is the principal contributor to the overall complexity, accounting for 82.73 G FLOPs and 96.74 M parameters. In comparison, integrating the AGLI module adds 26.34 G FLOPs and 38.21 M parameters but significantly enhances the model's ability to comprehend and process complex topology structures. Furthermore, the inclusion of the SGAED module increases the total complexity to 120.41 G FLOPs (up by 11.34 G) and 140.75 M parameters (up by 5.80 M). The SGAED module introduces a modest rise in computational complexity and parameters while greatly improving the model's performance in dealing with complex backgrounds. We also make a complexity comparison between our method and several state-of-the-art methods, which are reported in Table VII(b). Our method owns comparable FLOPs and parameters with ACCoNet [15] and EMFINet [47] and consistently outperforms them on all the datasets. Although our method owns higher complexity compared with other methods, our method also generally achieves a significant performance improvement (0.5%∼2% improvement) on these datasets.

E. Effect of Seed Selection on Performance

To examine the effect of the seed value selection on performance, we initialize the random number generator with five

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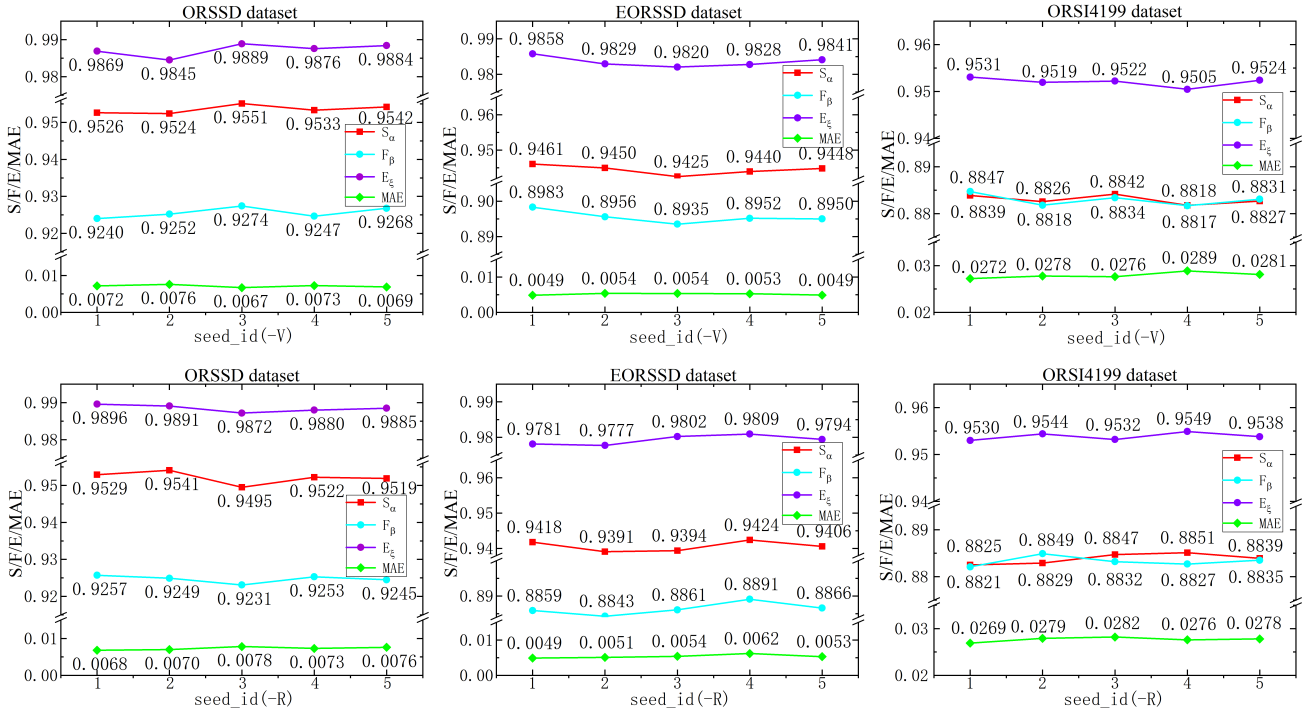


Fig. 9. Curves of S_α, F_β, E_ξ, and MAE on three benchmark datasets using different seed values. The top row shows results using the VGG-based model (Ours-V), while the bottom row shows results using the ResNet-based model (Ours-R).



Fig. 10. Some failure cases on the ORSI4199 dataset. (a) Optical RSI. (b) GT. (c) Ours.

different seed values and conduct the experiments under the same settings on three datasets. The experimental results are shown in Fig. 9, where it can be observed that the indicators exhibit slight fluctuations at a relatively high level with the change in seeds. We present a statistical analysis of the performance results in terms of mean ± standard deviation, as detailed below: on the ORSSD dataset, Ours-V generally achieves 0.9535 ± 0.0011 for S_α, 0.9256 ± 0.0014 for F_β, 0.9873 ± 0.0017 for E_ξ, and 0.0071 ± 0.0003 for MAE, while Ours-R generally achieves 0.9521 ± 0.0016, 0.9247 ± 0.001, 0.9885 ± 0.0009, and 0.0073 ± 0.0004, respectively. On the EORSSD dataset, Ours-V generally achieves 0.9445 ± 0.0013 for S_α, 0.8955 ± 0.0017 for F_β, 0.9835 ± 0.0014 for E_ξ, and 0.0052 ± 0.0002 for MAE, while Ours-R generally achieves 0.9407 ± 0.0014, 0.8864 ± 0.0017, 0.9793 ± 0.0013, and 0.0054 ± 0.0004, respectively. On the ORSI4199 dataset, Ours-V generally achieves 0.8830 ± 0.0009 for S_α, 0.8829 ± 0.0012 for F_β, 0.9520 ± 0.0009 for E_ξ, and 0.0279 ± 0.0006 for MAE, while Ours-R generally achieves 0.8838 ± 0.0011, 0.8833 ± 0.0010, 0.9539 ± 0.0007, and 0.0277 ± 0.0004, respectively. To maintain consistency with the performance representation of other methods, we finally report the mean values in Table I.

F. Failure Cases and Analysis

As previously mentioned, the proposed HFCNet demonstrates excellent capabilities in identifying and locating the salient objects in optical RSI. However, HFCNet still struggles with some particularly challenging scenes. As shown in the first two columns of Fig. 10, in low-contrast situations, the boundaries between the target and its surrounding environment are not clear, making it difficult for our model to distinguish between the target object and the background. Furthermore, as illustrated in the last two columns of Fig. 10, when an image contains objects with varied semantics, the model sometimes fails to detect the most salient object. This may be attributed to the uneven distribution of the training dataset and the limited semantic comprehension capabilities of the model. In the future, we will focus on designing a more effective RSI-SOD method to address these issues.

V. Conclusion

In this article, we delve into the application of SOD in the field of optical RSI and propose an HFCNet. The network consists of three key modules: the hybrid heterogeneous encoder, the AGLImodule, and the SGAED. The hybrid heterogeneous encoder uses CNN for local context modeling and incorporates transformer to assist in establishing long-range semantic relationships. The AGLI module can effectively integrate global semantic information and local detailed information through progressively aligning the heterogeneous features extracted by CNN and transformer. The proposed SGAED module uses deep salient cues to further filter shallow features, which

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suppresses irrelevant information and highlights important regions. Comprehensive experiments are conducted on three classical datasets, with the outcomes showcasing the superior performance of the proposed HFCNet.

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